

Sri Lanka Institute of Advanced Technological Education (SLIATE)

AI-POWERED LAPTOP MARKET VALUE PREDICTION & CAREER RECOMMENDATION SYSTEM

PROJECT REPORT



Advanced Technological Institute, Badulla

PROJECT REPORT PRESENTED BY: R.M.D.N.M. Rathnayaka

Registration No: BAD/IT/2324/F/050

Subject: Programming Individual Project

(HNDIT4052)

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DECLARATION

I do hereby declare that the work reported in this project report/thesis was exclusively carried out by me under the supervision of [Insert Supervisor's Name]. It describes the results of my own independent research except where due reference has been made in the text. No part of this project report/thesis has been submitted earlier or concurrently for the same or any other degree.

Date:

Certified by:

1. Supervisor (Name):

Signature:

ABSTRACT

Choosing a laptop that fits both a buyer's budget and their career path is difficult: comparison sites list specifications but rarely indicate whether a listed price is fair, and they offer no guidance on which specifications actually matter for a given profession. This project introduces the AI-Powered Laptop Market Value Prediction and Career Recommendation System, a web-based application that combines a machine-learning price predictor with a career-aware specification recommender behind a modern, responsive interface.

The system uses a Random Forest Regression model, built with scikit-learn and trained on a curated CSV dataset of laptop listings using pandas and numpy for data preparation, to estimate the current market value of a laptop from its specifications. The trained model is persisted with pickle and served through a lightweight Python backend (http.server and socketserver) that exposes prediction results to a React-based frontend. Alongside the price estimate, a career-recommendation engine maps the user's stated career or workflow to a prioritised set of recommended specifications, helping students and professionals judge whether a listing represents good value and whether it suits their intended use.

The resulting prototype allows a user to enter laptop specifications and a target career, receive an explainable price prediction and specification recommendation, and export the results for later comparison. [Insert a summary of your evaluation results here, e.g. model accuracy/ R^2 score and any user testing outcomes.]

Title of the Project: AI-Powered Laptop Market Value Prediction and Career Recommendation System

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R.M.D.N.M. Rathnayaka.

TABLE OF CONTENTS

DECLARATION.....	II
ABSTRACT.....	III
ACKNOWLEDGEMENTS	IV
LIST OF FIGURES	V
LIST OF TABLES	VI
LIST OF ABBREVIATIONS	VII
CHAPTER 1: INTRODUCTION.....	1
1.1 Problem Statement.....	1
1.2 Project Objectives.....	2
1.3 Functional Requirements.....	2
CHAPTER 2: LITERATURE REVIEW.....	3
2.1 Existing Laptop Comparison Tools.....	3
2.2 Machine Learning for Price Prediction	3
2.3 Motivation	4
CHAPTER 3: MATERIALS AND METHODS	5
3.1 Materials (Hardware & Software)	5
3.2 Dataset	6
3.3 Methodology.....	6
CHAPTER 4: TECHNOLOGIES & SYSTEM DESIGN	8
4.1 Frontend and Backend Stack	8
4.2 Machine Learning Pipeline.....	8
4.3 System Architecture	9
4.4 Runtime / Request Flow	10
4.5 Project Schedule & Budget Summary	11
CHAPTER 5: RESULTS AND DISCUSSION	13
5.1 Objectives Achieved.....	13
5.2 Verification & Evaluation Results.....	13
5.3 Limitations.....	14
5.4 Future Development	14
CHAPTER 6: CONCLUSIONS	15
REFERENCES.....	16
APPENDIX A: ADDITIONAL SCREENSHOTS	17

LIST OF FIGURES

Figure 1: System Architecture Overview..... 9
Figure 2: Runtime / Request Flow Diagram..... 10
Figure 3: Project Schedule Gantt Chart..... 12

LIST OF TABLES

Table 1: Server and Client Hardware Requirements..... 5
Table 2: System Development Software Requirements..... 6
Table 3: Technology Stack by Layer..... 8
Table 4: Project Schedule Table 11
Table 5: Project Budget Summary Table 12

LIST OF ABBREVIATIONS

- API: Application Programming Interface
- ATI: Advanced Technological Institute
- CSV: Comma-Separated Values
- HND: Higher National Diploma
- HTML: Hypertext Markup Language
- JSON: JavaScript Object Notation
- LKR: Sri Lankan Rupee
- ML: Machine Learning
- RAM: Random Access Memory
- RF: Random Forest
- SDLC: Software Development Life Cycle
- SLIATE: Sri Lanka Institute of Advanced Technological Education
- UI: User Interface

CHAPTER 1: INTRODUCTION

The Sri Lanka Institute of Advanced Technological Education (SLIATE) is a leading national institution established to provide Higher National Diploma (HND) education. As technology continues to reshape local job markets, students and professionals are increasingly reliant on personal computing devices suited to their specific field of work. The AI-Powered Laptop Market Value Prediction and Career Recommendation System was developed to help this audience make informed, career-aligned laptop purchasing decisions.

This project combines machine learning price prediction with a career-aware recommendation engine and a modern web interface, delivering easy, actionable guidance to buyers who would otherwise have to rely on scattered specification sheets and guesswork.

1.1 Problem Statement

Two related problems motivate this project:

- Decision overload: students and professionals struggle to choose the right laptop tailored to their career goals, as most comparison sites list specifications but do not connect those specifications to real workflows.
- Uncertain pricing: buyers find it hard to estimate the current market value of a used or new laptop before purchase, leaving them exposed to overpaying or missing a fair deal.

Existing comparison platforms leave a clear gap: they surface specifications but do not offer predictive pricing or career-driven recommendations in one place.

1.2 Project Objectives

- Predict laptop market price using AI, producing accurate, explainable estimates.
- Recommend laptop specifications tailored to specific careers and workflows.
- Deliver a modern, responsive web interface for interactive use.
- Allow export of prediction results for comparison and record-keeping.
- Improve purchasing decisions by combining price and career fit in a single tool.

1.3 Functional Requirements

- FR1 (Specification Input): Allow the user to enter laptop specifications (e.g. brand, processor, RAM, storage, GPU, screen size) through the web interface.
- FR2 (Price Prediction): Compute an estimated market price from the entered specifications using the trained Random Forest regression model.
- FR3 (Career Input): Allow the user to select or enter a target career or workflow.
- FR4 (Career-Aware Recommendation): Return a prioritised set of recommended specifications suited to the selected career.
- FR5 (Results Display): Present the predicted price and recommendations in a clear, responsive React interface.
- FR6 (Export): Allow the user to export prediction results for later comparison and record-keeping.

- FR7 (Model Serving): Serve the persisted model through a backend API built with Python's `http.server` and `socketserver`.

PLACEHOLDER — Review these functional requirements against your actual implementation and add/remove items as needed.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Laptop Comparison Tools

A range of laptop comparison and shopping-assistant websites already exist, letting users filter devices by specification and compare listed prices side by side. However, these platforms are typically descriptive rather than predictive: they show what a device is and what it currently costs, without estimating whether that price reflects fair market value or degrades over time. Few, if any, connect specification choices to specific careers or workflows, leaving buyers to translate generic spec sheets into a purchasing decision on their own.

2.2 Machine Learning for Price Prediction

Regression-based machine learning models are well established for price-estimation problems where a numeric target (such as market price) depends on a mix of categorical and numeric features (such as brand, processor, RAM and storage). Random Forest Regression, an ensemble of decision trees, is a common choice for this kind of tabular data because it handles non-linear relationships between specifications and price without requiring extensive manual feature engineering, and it remains reasonably interpretable through feature-importance analysis.

PLACEHOLDER — Add citations to the specific papers, articles, or datasets you referenced for your literature review, and expand this section with a comparison of any similar academic or commercial systems you studied.

2.3 Motivation

The development of this system was motivated by the absence of a single tool that unifies predictive pricing with career-driven specification guidance. Combining a Random Forest price predictor with a career-aware recommendation engine, behind one responsive interface, helps students and professionals bridge the gap between a generic specification sheet and a confident purchasing decision.

CHAPTER 3: MATERIALS AND METHODS

3.1 Materials (Hardware & Software)

Hardware Requirements

Component	Specification
Development / Client PC	[Insert CPU] / [Insert RAM] / [Insert free storage]
Server (if deployed)	[Insert server specification]

Table 1: Server and Client Hardware Requirements

Software Requirements

Layer	Software / Version
Operating System	[Insert OS used, e.g. Windows 11 / Ubuntu 22.04 LTS]
Language / Runtime	Python 3.x
Backend Server	Python http.server & socketserver
Frontend	React JS
Machine Learning	scikit-learn (Random Forest Regression)
Data Handling	pandas, numpy
Model Persistence	pickle
Data Storage	CSV dataset
Development Tools	Jupyter Notebook, [Insert code editor / IDE, e.g. VS Code], Git

Table 2: System Development Software Requirements

3.2 Dataset

The model was trained on a CSV dataset of laptop listings covering common specification fields (for example brand, processor, RAM, storage type and capacity, GPU and screen size) alongside a recorded market price. pandas and numpy were used to clean and prepare this data before training. [Insert the exact source, size, and feature list of your dataset, and describe any cleaning or feature-engineering steps you applied.]

3.3 Methodology

The project followed a structured, phase-by-phase development process:

[Data Collection] → [Data Preprocessing] → [Model Training] → [Model Evaluation] → [Backend API Development] → [Frontend Development] → [Integration & Testing] → [Deployment]

- Data Collection: Gathered a CSV dataset of laptop listings with specifications and market prices.
- Data Preprocessing: Cleaned and transformed the dataset using pandas and numpy, preparing categorical and numeric features for training.

- **Model Training:** Trained a Random Forest Regression model with scikit-learn in Jupyter Notebook to predict market price from specifications.
- **Model Evaluation:** Assessed prediction accuracy on held-out data and persisted the final model with pickle.
- **Backend API Development:** Built a Python backend using `http.server` and `socketserver` to load the trained model and serve predictions.
- **Frontend Development:** Built a responsive, component-driven React interface for entering specifications and viewing results.
- **Integration & Testing:** Connected the React frontend to the backend API and verified the end-to-end prediction and recommendation flow.
- **Deployment:** [Describe how/where the system was deployed, or mark as a local prototype.]

CHAPTER 4: TECHNOLOGIES & SYSTEM DESIGN

4.1 Frontend and Backend Stack

Layer	Technology
Presentation (Frontend)	React JS — responsive UI, component-driven design
Application Server (Backend)	Python http.server + socketserver — custom web server serving the React frontend and model predictions
Machine Learning	Random Forest Regression via scikit-learn, trained in Jupyter Notebook
Data Preparation	pandas / numpy
Model Persistence	pickle
Data Storage	CSV dataset for model training and quick iteration

Table 3: Technology Stack by Layer

4.2 Machine Learning Pipeline

The CSV dataset is loaded and cleaned with pandas, with numeric and categorical specification fields prepared as model features using pandas/numpy. A Random Forest Regression model is trained on this feature set with scikit-learn inside Jupyter Notebook, then persisted to disk with pickle so it can be loaded by the backend without retraining at request time. The career-recommendation engine maps a selected career to a prioritised set of specification weightings, which is combined with the price prediction when the frontend requests a recommendation.

4.3 System Architecture

The system follows a modular pipeline: a React frontend collects user inputs and displays results. The inputs are sent to a Python http.server backend, which loads the trained AI model. The model returns a price prediction and recommended specifications, and the backend sends the results back to the frontend.

*[React Frontend] → [Python http.server Backend] → [Persisted Random Forest Model] →
[Prediction + Recommendation] → [Frontend Display]*

Figure 1: System Architecture Overview

4.4 Runtime / Request Flow

The runtime flow separates concerns clearly, which supports easy testing, scaling, and model updates:

- Input Select — the user enters laptop specifications and a target career in the React interface.
- Send Request — the frontend sends the input as a request to the backend API.
- Preprocess — the backend prepares the incoming data to match the model's expected feature format.
- Model Inference — the persisted Random Forest model produces a price prediction; the recommendation engine produces career-matched specifications.

- Display & Export — the results are returned to the frontend for display, with an option to export them for comparison.

[Input Select] → [Send Request] → [Preprocess] → [Model Inference] → [Display & Export]

Figure 2: Runtime / Request Flow Diagram

4.5 Project Schedule & Budget Summary

Project Schedule Table

Milestone / Phase	Duration	Start Date	End Date
Data Collection & Preprocessing	[Insert]	[Insert]	[Insert]
Model Training & Evaluation	[Insert]	[Insert]	[Insert]
Backend API Development	[Insert]	[Insert]	[Insert]
Frontend Development	[Insert]	[Insert]	[Insert]
Integration & Testing	[Insert]	[Insert]	[Insert]
Deployment & Documentation	[Insert]	[Insert]	[Insert]

Table 4: Project Schedule Table

PLACEHOLDER — Insert your actual project timeline, and add a Gantt chart image here as Figure 3 if you have one.

Project Budget Table (LKR)

Item Group	Details	Cost (LKR)
Development	[Insert]	[Insert]
Hosting & Setup	[Insert]	[Insert]
Total Cost		[Insert]

Table 5: Project Budget Summary Table

PLACEHOLDER — Insert your actual cost breakdown, or remove this table if the project had no associated budget.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Objectives Achieved

- Price Prediction Model: Trained a Random Forest regression model producing market price predictions, with evaluation metrics recorded during validation.
- Career-Aware Recommendation Engine: Implemented a recommendation engine mapping careers to prioritised specifications.
- Interactive Prototype: Delivered a React + Python http.server prototype with an interactive UI and a backend API that serves the AI model and returns predictions.
- Export Functionality: Allowed prediction results to be exported for comparison and record-keeping.
- Documentation: Produced system design, dataset description, model training, and deployment notes.

5.2 Verification & Evaluation Results

PLACEHOLDER — Insert your model's evaluation metrics here (e.g. R^2 , MAE, RMSE on a held-out test set), any screenshots of predictions compared against real listings, and a short discussion of how well the price predictions and career recommendations matched expectations.

5.3 Limitations

- Dataset Coverage: A CSV-based dataset of finite size limits the model's ability to generalise to brands, regions, or configurations not well represented in training data.
- Non-Production Backend: Python's http.server and socketserver are lightweight and well suited to a prototype, but are not designed for high-concurrency production traffic.
- Single-Model Approach: Price prediction relies on one Random Forest model; no ensemble or alternative model comparison was performed.
- Recommendation Granularity: Career-to-specification mappings are based on a fixed set of career profiles and may not capture every individual workflow.

PLACEHOLDER — Review and adjust these limitations to match what you actually observed while building and testing the system.

5.4 Future Development

- Expand the training dataset with a larger and more diverse set of laptop listings to improve prediction accuracy.
- Migrate the backend to a production-grade framework (e.g. Flask, FastAPI, or Django) for improved scalability and security.
- Add user accounts to save and revisit past predictions and comparisons.
- Extend the recommendation engine with more granular, user-customisable career profiles.

CHAPTER 6: CONCLUSIONS

The AI-Powered Laptop Market Value Prediction and Career Recommendation System successfully combines a Random Forest regression price predictor with a career-aware recommendation engine behind a responsive React interface. By pairing an explainable price estimate with specification guidance tailored to the user's career, the system reduces decision overload and pricing uncertainty for students and professionals shopping for a laptop. The React and Python `http.server` prototype demonstrates that a lightweight, modular pipeline — input, API, preprocessing, model inference, and results/export — can deliver an interactive, useful tool without requiring heavyweight infrastructure.

REFERENCES

PLACEHOLDER — Replace the entries below with the actual sources you referenced (papers, documentation, datasets, or tools) while building this project.

1. [Author], "[Title of source on laptop price prediction / Random Forest regression]," [Journal / Conference / Website], [Year].
2. scikit-learn developers, "Random Forest Regressor — scikit-learn documentation," scikit-learn.org.
3. React core team, "React Documentation," react.dev.
4. Python Software Foundation, "http.server — HTTP servers," docs.python.org.

APPENDIX A: ADDITIONAL SCREENSHOTS

Prediction Interface

[Insert a screenshot of the React interface where the user enters specifications and career, and views the predicted price and recommendations.]

PLACEHOLDER — Insert screenshot here and label it Figure A1.

Recommendation Results

[Insert a screenshot showing the career-aware specification recommendations and any export functionality.]

PLACEHOLDER — Insert screenshot here and label it Figure A2.